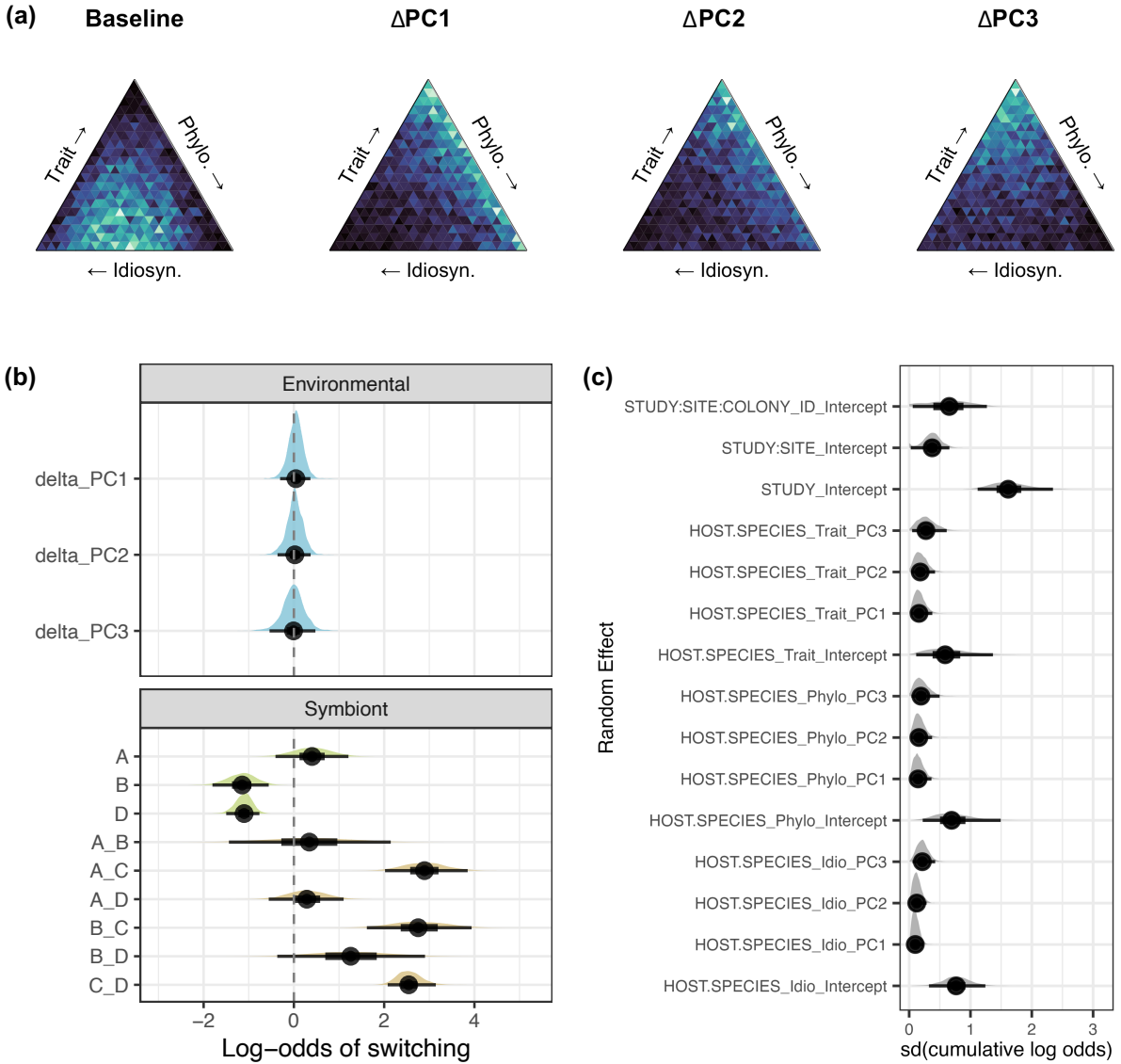

(a)
Baseline
ΔPC1
ΔPC2
ΔPC3
Trait →
Phylo. →
← Idiosyn.
(b)
Environmental
delta_PC1
delta_PC2
delta_PC3
Symbiont
A
B
D
A_B
A_C
A_D
B_C
B_D
C_D
−2
0
2
4
Log−odds of switching
(c)
STUDY:SITE:COLONY_ID_Intercept
STUDY:SITE_Intercept
STUDY_Intercept
HOST.SPECIES_Trait_PC3
HOST.SPECIES_Trait_PC2
HOST.SPECIES_Trait_PC1
HOST.SPECIES_Trait_Intercept
HOST.SPECIES_Phylo_PC3
HOST.SPECIES_Phylo_PC2
HOST.SPECIES_Phylo_PC1
HOST.SPECIES_Phylo_Intercept
HOST.SPECIES_Idio_PC3
HOST.SPECIES_Idio_PC2
HOST.SPECIES_Idio_PC1
HOST.SPECIES_Idio_Intercept
Random Effect
0
1
2
3
sd(cumulative log odds)